3 **(a)** **(i)** Define *gravitational potential* at a point in a gravitational field.

……

……

…… [1]

(ii) Explain why gravitational potential is always negative.

……

……

……

……

…… [2]

(b) Fig. 3.1 shows the variation of gravitational potential between the surface of Moon and the surface of Earth along the line joining their centres.

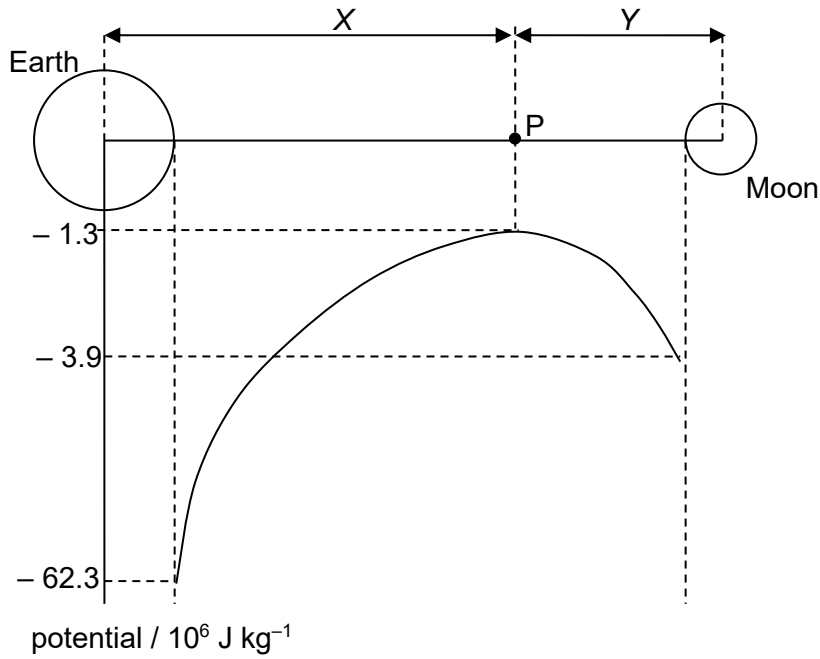

Fig. 3.1

Explain why the gradients of the graph near the surface of Earth and that near the surface of Moon have opposite signs.

……

……

……

…… [2]

(c) Two moons A and B move in circular orbits about a planet, as illustrated in Fig. 3.2.

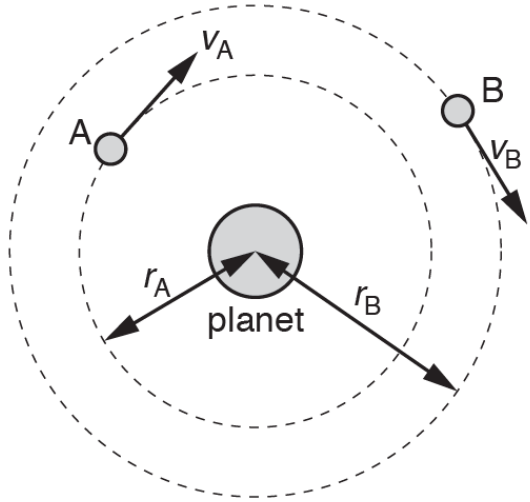

Fig. 3.2

Moon A has an orbital radius r_A of 1.3×10^8 m, linear speed v_A and orbital period T_A.
Moon B has an orbital radius r_B of 2.2×10^{10} m, linear speed v_B and orbital period T_B.

(i) Determine the ratio $\frac{v_A}{v_B}$

ratio = …… [2]

(ii) Show that $\frac{T_A}{T_B} = 4.5 \times 10^{-4}$.

[1]

(iii) The planet spins about its own axis with angular speed 1.7×10^{-4} rad s^{-1}. Moon A is always above the same point on the planet's surface. Determine the orbital period T_B of moon B.

T_B = ... s [2]